

Internship Project

E-Commerce Cohort Retention & Customer Lifetime Value Analysis

Understanding Customer Retention, Churn Patterns & Lifetime Value through Cohort and Retention Analytics

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Project Domain : E-commerce Cohort Retention and CLTV Analysis

Tools Stack : Python (pandas, numpy, matplotlib), DAX, PowerBI, GitHub

Executive Problem Statement

Business Challenge

Acquiring new customers is significantly more expensive than retaining existing ones. While many businesses focus on customer acquisition, they often overlook how long customers stay engaged and how much value they generate over time. Without understanding customer retention and lifetime value, organizations risk increasing acquisition costs while experiencing declining profitability.

Project Goal

This project applies Cohort Analysis and Customer Lifetime Value (CLTV) Analysis to evaluate customer behavior after acquisition. By grouping customers based on their first purchase month, the analysis identifies retention trends, measures customer value, and uncovers opportunities to improve long-term business growth.

Dataset Overview

Aspect	Details
Data Type	E-commerce Transactional Data
Analysis Period	Oct 2016 - Aug 2018
Granularity	Individual Customer Transactions
Key Entity	Customer
Core Measures	Revenue, Orders, Customers
Customer Analysis	Cohort, Retention, Segemnt, CLTV

Tables used

1. Olist customers dataset
2. Olist order items
3. Olist order payments
4. Olist Orders dataset
5. Olist products
6. Product category name

Key Metrics Derived

1. Historical CLTV
2. Purchase Frequency
3. Average Order Value
4. Customer Segmentation
5. Cohort Months
6. Retention Rate
7. Total Revenue, Customers, Orders

Business Questions
Answered

01

When and how quickly do customers churn after acquisition?

02

What purchasing behaviors drive higher customer value?

03

Which customer cohorts have the strongest retention?

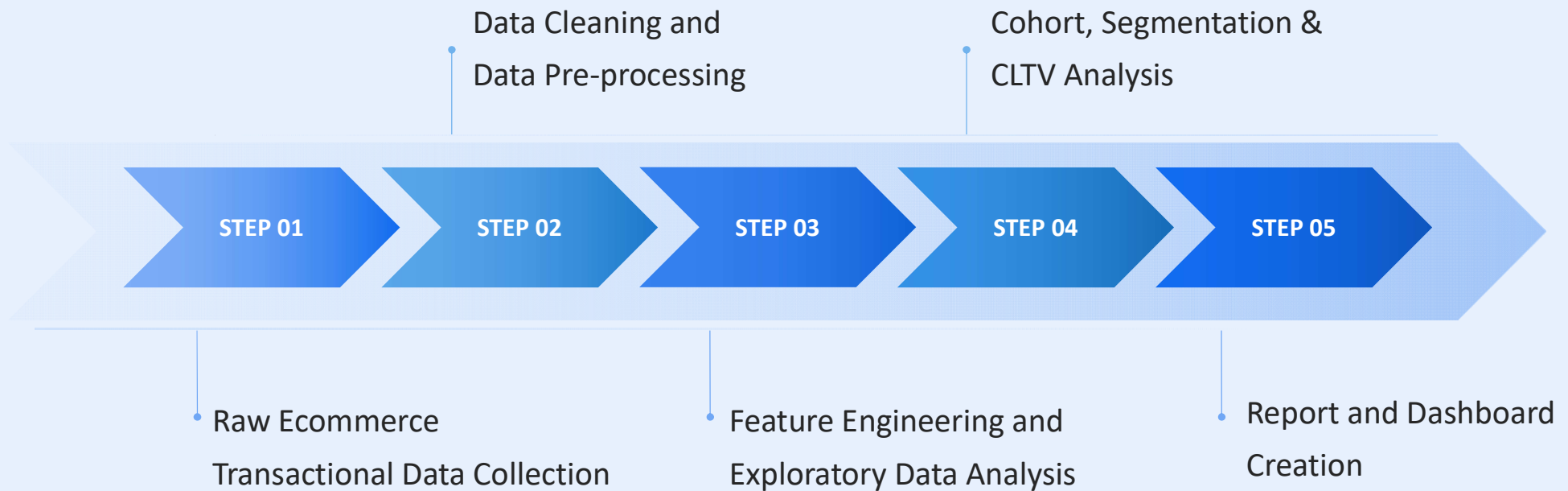
04

Which products and customer segments contribute most to revenue and value?

05

Which customer segments generate the highest CLTV?

Analytical Pipeline



Cohort Retention Matrix

Objective: Understand how customer retention changes after the first purchase and identify major drop-off points.

Cohort	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	19	20
Dec 2016	100%	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Jan 2017	100%	28%	28%	14%	42%	14%	42%	14%	14%	0%	42%	14%	70%	42%	14%	14%	28%	42%	14%	0%
Feb 2017	100%	18%	31%	12%	43%	12%	25%	18%	12%	18%	12%	31%	12%	18%	12%	6%	6%	18%	0%	0%
Mar 2017	100%	44%	36%	40%	36%	16%	16%	32%	32%	8%	36%	12%	20%	12%	16%	24%	8%	12%	0%	0%
Apr 2017	100%	62%	22%	18%	27%	27%	35%	31%	31%	18%	27%	9%	4%	4%	9%	9%	13%	0%	0%	0%
May 2017	100%	46%	46%	29%	29%	32%	41%	14%	26%	26%	26%	35%	23%	3%	17%	20%	0%	0%	0%	0%
Jun 2017	100%	49%	40%	43%	30%	40%	36%	23%	13%	20%	30%	36%	16%	16%	23%	0%	0%	0%	0%	0%
Jul 2017	100%	53%	35%	24%	29%	21%	32%	11%	19%	27%	21%	29%	13%	24%	0%	0%	0%	0%	0%	0%
Aug 2017	100%	69%	35%	27%	35%	52%	30%	27%	15%	15%	25%	20%	12%	0%	0%	0%	0%	0%	0%	0%
Sep 2017	100%	70%	55%	27%	45%	22%	22%	25%	27%	17%	25%	7%	0%	0%	0%	0%	0%	0%	0%	0%
Oct 2017	100%	72%	25%	9%	23%	21%	21%	37%	28%	18%	21%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Nov 2017	100%	57%	37%	17%	17%	18%	11%	18%	13%	6%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Dec 2017	100%	21%	28%	34%	26%	21%	17%	2%	19%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Jan 2018	100%	34%	37%	29%	29%	16%	18%	23%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Feb 2018	100%	35%	40%	30%	25%	22%	21%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Mar 2018	100%	40%	30%	30%	12%	12%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Apr 2018	100%	59%	30%	24%	14%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
May 2018	100%	52%	26%	18%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Jun 2018	100%	43%	27%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Jul 2018	100%	52%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Aug 2018	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%

Key Metrics to Highlight

- Month 0: Initial customer cohort
- Month 1, Month 2, ...: Subsequent customer activity
- Retention %: Percentage of the original cohort remaining active

Key Insight

Customer retention declines sharply after the initial purchase, with the largest customer drop-off occurring in the early months following acquisition.

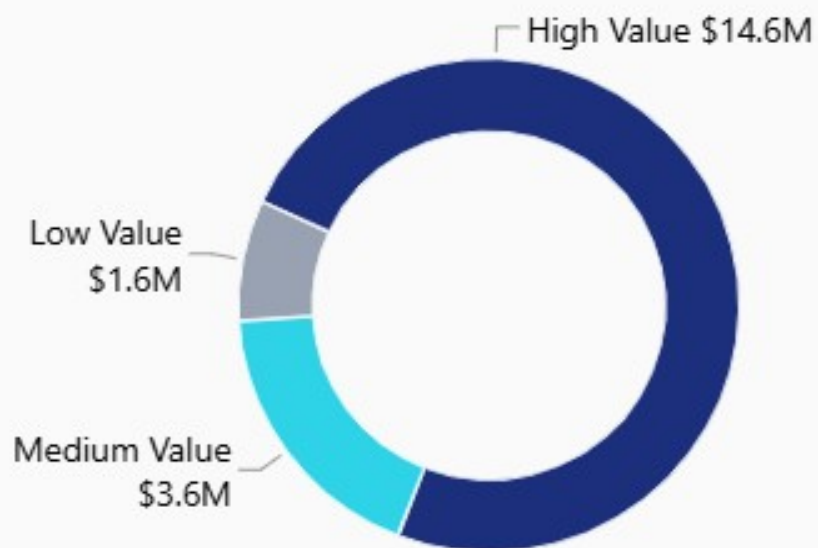
Business Implication

Early post-purchase engagement is critical. Targeted re-engagement campaigns during the first few months could help reduce customer churn.

Customer Value Segmentation

Objective: Identify high-value, medium, and low-value customers based on their purchasing behavior and CLTV.

Customer Segment Summary



Segmentation

Customers from the data were segmented into High, Medium, and Low categories based on their Historical CLTV using `pd.qcut()` function in Python where each category had around 31k customers.

Key Insight

High-value customers represent a similar-sized customer group but generate substantially greater customer value than the lower-value segments.

Business Implication

High-value customers should be prioritized for retention initiatives, personalized offers, and Customer loyalty programs.

Customer Lifetime Value Analysis

Objective: Quantify the long-term economic value generated by each customer.

Key steps in calculating Historical CLTV :

1. Deriving Customer Level summary
2. Calculating customer lifespan
3. Calculating average order value for each customer
4. Calculating purchase frequency based on orders
5. Calculating Historical CLTV

Key Findings

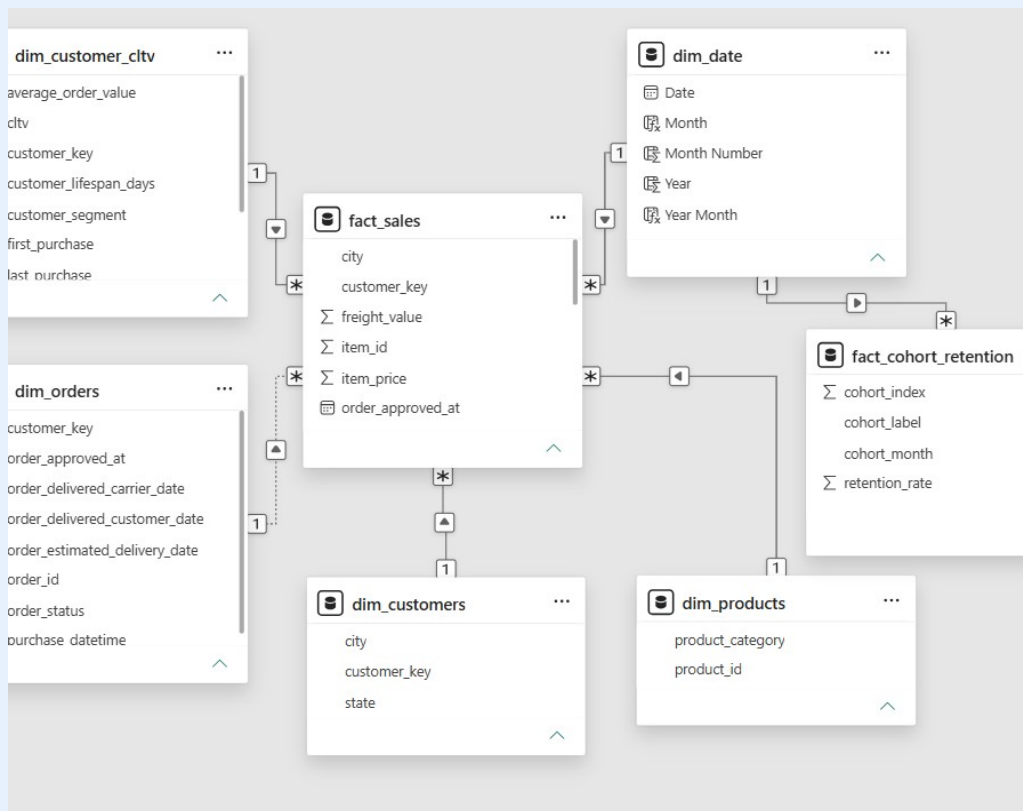
- CLTV varies substantially across customer segments.
- Higher-value customers demonstrate stronger purchasing behavior.
- AOV and purchase frequency are important contributors to customer value.

Business Implication

Increasing repeat purchases and extending customer lifespan can significantly improve customer lifetime value.

Data Modeling & Analytical Architecture

Objective: Building a structured data model that connects transactional data, customer data , cohort, and date information for reliable analysis and interactive reporting.



Fact Tables

- *fact_sales* : transactional level sales data
- *fact_cohort_retention* : customer cohort data

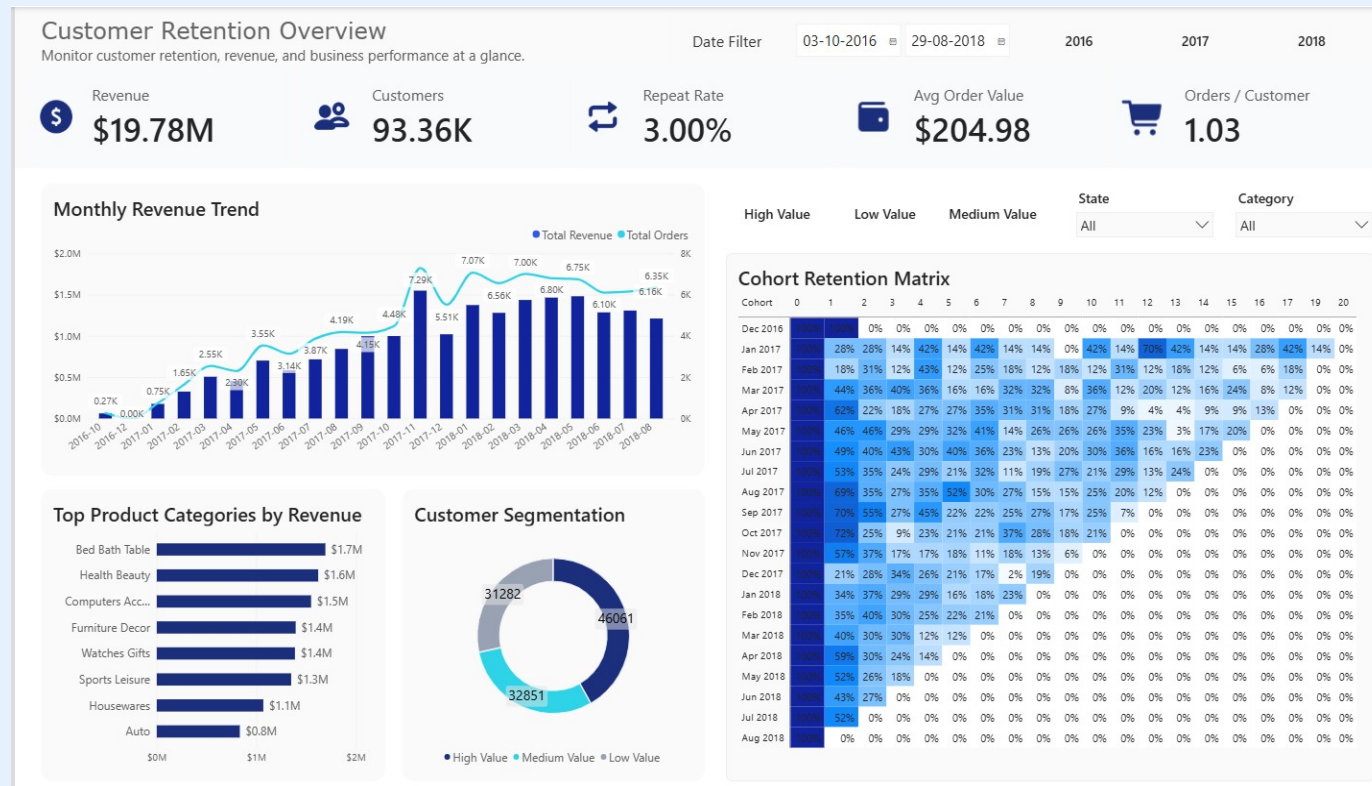
Dimension tables

- *dim_customers* : customer attributes
- *dim_customer_cltv* : customer-level CLTV metrics
- *dim_date* : calendar/date analysis
- *dim_orders* : order level data
- *dim_products* : product Information

Relationships

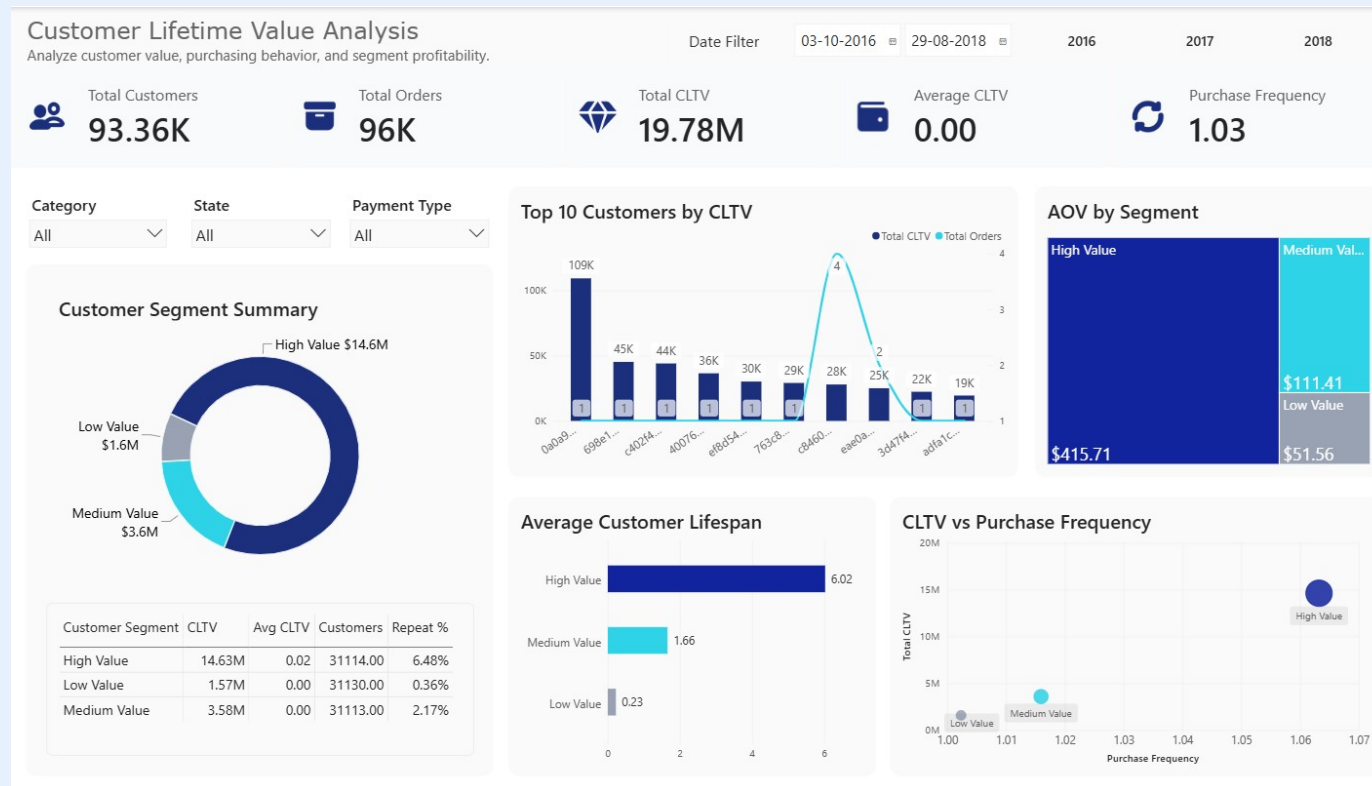
One-to-many relationships connect dimension tables to the relevant fact tables, allowing filters such as customer segment, date, and cohort to flow into the analytical visuals.

Customer Retention Overview Dashboard



Description : The dashboards contains key business kpi's along with monthly revenue trend, cohort retention matrix, top product categories and customer segmentation chart.

Customer Lifetime Value & Customer Segmentation



Description : The dashboards contains key CLTV kpi's like Total CLTV, Avg CLTV and Purchase Frequency along with chart visuals like top customers, AOV by segment, AVG lifespan bar chart, CLTV vs Purchase Frequency distribution, Segmentation summary matrix etc.

Key Insights Derived

1. Revenue is strong, but repeat purchasing is low

The business generated **\$19.78M** revenue from **~96K** orders and **93.36K** customers, yet the overall repeat rate is only **3%** and orders/customer is just **1.03**.

Insight: Revenue is being driven largely by first-time/low-frequency purchases rather than strong repeat behaviour.

2. High-value customers are the major revenue driver

The High Value segment contributes **\$14.63M** CLTV, compared with **\$3.58M** for Medium Value and **\$1.57M** for Low Value.

Insight: High-value customers are important to business revenue and should receive the strongest retention and loyalty efforts.

3. Customer value differs dramatically across segments

High Value customers have an AOV of **\$415.71**, versus **\$111.41** for Medium Value and only **\$51.56** for Low Value. Their average lifespan is also much longer (**6.02** vs. 1.66 vs. 0.23).

Insight: Increasing customers' order value and keeping them active longer appears to be the key path to moving customers toward the High Value segment.

Key Insights Derived

4. Cohort retention declines sharply over time

The cohort matrix shows that customers generally experience a large drop in retention after their initial purchase, with many cohorts falling into low double-digit retention or below within subsequent periods.

Insight: The major challenge is not acquisition but converting first-time buyers into repeat customers.

5. Some cohorts demonstrate better retention than others

The cohort matrix shows noticeable differences between acquisition cohorts. For example, several 2017 cohorts maintain stronger retention in their early periods than others.

Insight: The business should investigate what acquisition channels, products, promotions, or customer experiences were associated with stronger cohorts and replicate those approaches.

6. Customer value is highly concentrated at the top

The Top 10 CLTV visual shows a very large gap between the highest-value customer (~\$109K) and the remaining top customers, while the CLTV vs Purchase Frequency visual also shows the High Value segment clearly separated from Medium and Low Value customers.

Insight: A relatively small group of highly valuable customers contributes disproportionately to lifetime value, making VIP/loyalty programs and personalized retention strategies commercially important.

Business Recommendations

01

Strengthen Early Retention

Target new customers with personalized offers and follow-up campaigns during the first few months, where cohort retention drops sharply.

02

Build a High-Value Customer Program

Prioritize High Value customers with loyalty rewards, exclusive offers, and personalized communication to protect the \$14.63M CLTV contribution.

03

Increase Repeat Purchases

With only a 3% repeat rate and 1.03 orders/customer, introduce post-purchase campaigns, replenishment reminders, and cross-selling strategies.

04

Increase Customer Value

Use bundles, cross-selling and premium recommendations to increase AOV, particularly by moving Medium Value customers toward the High Value segment.

05

Replicate High-Performing Cohorts

Identify the products, categories, acquisition periods, and customer characteristics associated with stronger-retaining cohorts and use them to improve future acquisition campaigns.



Conclusion and Key Takeaways

01 — Strong Revenue Base

The business generated \$19.78M revenue, providing a strong foundation for customer-value growth.

02 — Retention is the Main Opportunity

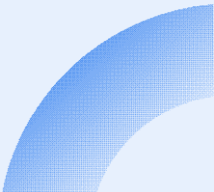
The 3% repeat rate and declining cohort retention indicate that converting first-time customers into repeat buyers is a major challenge.

03 — High-Value Customers Drive CLTV

The High Value segment contributes \$14.63M CLTV, with substantially higher AOV and customer lifespan than the other segments.

04 — Focus on Customer Lifetime Value

Improving retention, purchase frequency, customer lifespan, and AOV can increase long-term customer profitability.





Thank You

E-commerce Cohort Retention & CLTV Analytics Project